

|         |                                         |                                       |                                               |
|---------|-----------------------------------------|---------------------------------------|-----------------------------------------------|
| 5 years | 1,800 +                                 | 386                                   | 10,434                                        |
| in IT   | customer cases closed<br>complaint-free | regression assertions in FlowGT<br>CI | repair cases in RepairIQ v2<br>knowledge base |

Summary

Christchurch-based AI engineer building agentic applications with Claude, OpenAI and Qwen: tool calling, planning and evaluation, integrated with third-party APIs and shipped with observability. Master of Applied Computing graduate with five years in IT. I use AI coding agents daily and review their output like a lead reviewer: specs in, judgment out, and I own the merge.

Career

Oct 2019 – Present · Ningbo → PB Tech → AI projects: three roles, five years in IT, now shipping AI applications with Claude, OpenAI and Qwen.

|      |                                                                                                              |
|------|--------------------------------------------------------------------------------------------------------------|
| 2019 | Ningbo Yuanyuan: technical lead for a five-person Alipay Mini Program platform.                              |
| 2024 | PB Tech: service technician in Christchurch; 1,800+ cases; retail-robot automation.                          |
| 2026 | AI projects: a live RAG repair-diagnosis agent, a serverless job-matching platform and LLM-assisted tooling. |

Experience

Service Technician · PB Tech · Christchurch

Oct 2024 – Present

|         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ROLE    | Service technician at New Zealand's largest IT retailer, owning faults end to end.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| ACTIONS | <ul style="list-style-type: none"><li>Integrated the store's Pudu retail robot with its cloud API (HMAC-SHA1 signed requests); built a Python pipeline: Playwright scrape → 1080 × 1920 ad frames with scan-to-buy QR codes → human review → upload; packaged the workflow as a Claude Code agent skill</li><li>Structured a 500+ entry product and after-sales knowledge base, now live on the robot as a customer-facing shop assistant</li><li>Diagnose Windows, macOS, Microsoft 365 and hardware faults to component level; handle vendor RMAs and out-of-warranty quotes through to closure</li><li>Handle front-counter intake and the service phone line daily in English and Mandarin</li><li>Launched the first 44 promotional ads on the robot and mapped products to 28 shelf points</li></ul> |
| IMPACT  | <ul style="list-style-type: none"><li>1,800+ customer cases closed complaint-free against agreed response times</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

Technical Lead · Ningbo Yuanyuan E-commerce · Ningbo, China

Oct 2019 – Oct 2022

|         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ROLE    | Technology lead for a five-person device-rental and instalment platform on Alipay.                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| ACTIONS | <ul style="list-style-type: none"><li>Led and personally built the Alipay Mini Program platform — front-end pages, back-end APIs and database schema — with one developer covering separate modules</li><li>Set up the release pipeline and automated testing; ran Linux servers with patching, monitoring and on-call response</li><li>Built the operational data layer with SQL extraction, Python analysis and Tableau dashboards to set regional ad targeting</li><li>Served 500+ instalment customers across 20,000+ transactions</li></ul> |
| IMPACT  | <ul style="list-style-type: none"><li>Cut advertising cost by 50% while doubling order volume via regional targeting</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                   |

Technical Support Volunteer · TechMate · Christchurch

Mar 2024 – Sep 2024

|         |                                                                                                                                                   |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| ROLE    | Volunteer at weekly drop-in sessions where older residents arrived with no fault description.                                                     |
| ACTIONS | <ul style="list-style-type: none"><li>Listened to each resident's description, then had them work the device to find the sticking point</li></ul> |
| IMPACT  | <ul style="list-style-type: none"><li>Supported 50+ older residents one to one over six months of sessions</li></ul>                              |

Projects

|                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RepairIQ — AI repair-diagnosis agent                | Live (v1) · Jan 2026 – Present                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| STACK                                               | Python · Claude API / Agent SDK · OpenAI API · Docker · Render; v2 on Node.js / Express / SQLite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| BUILT                                               | <ul style="list-style-type: none"><li>• Question-driven agent that narrows down faults and returns fix steps, a "needs a professional" call or a cost range</li><li>• RAG over iFixit repair guides and manufacturer docs: embeddings, vector search and custom tool calls (v1)</li><li>• Evaluated answers against test question sets; price ranges compiled from industry quote experience</li><li>• v2 rewrite: <a href="#">10,434</a> repair cases in SQLite feeding a chain-of-thought JSON prompt; provider-agnostic Qwen / OpenAI / Gemini layer with SSE streaming and fallback; <a href="#">75</a> automated tests</li></ul> |
| Job Search Workbench — LLM-assisted job-search tool | Personal, used daily · 2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| STACK                                               | Next.js <a href="#">16</a> · React <a href="#">19</a> · TypeScript · Prisma · SQLite · Zod · Vitest · Playwright · Claude Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| BUILT                                               | <ul style="list-style-type: none"><li>• Runs Claude headless with tools disabled: minimal prompts, batching, retries and timeouts; logs tokens and cost per call</li><li>• Typed judgments from the TypeSafe Jev API validated with Zod; uncertain results go to human review</li><li>• Analyses job descriptions against my profile and generates tailored CVs and cover letters from a fixed template</li></ul>                                                                                                                                                                                                                     |
| FlowGT — Full-Stack Developer (Volunteer)           | Live · Jan 2026 – Present                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| STACK                                               | JavaScript · Cloudflare Pages Functions / Workers / D1 / Workers AI (Qwen3-30B) · GitHub Actions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| BUILT                                               | <ul style="list-style-type: none"><li>• In a 3–5 person volunteer team, built much of a live serverless job-matching platform: <a href="#">35</a> APIs and a <a href="#">35</a>-table schema</li><li>• LLM job-description analyser on Workers AI: JSON-Schema-constrained output, automatic retry, SHA-256 input-hash caching</li><li>• AI spend capped by a quota counted before inference and refunded on failure, with tiered daily limits</li><li>• <a href="#">386</a>-assertion regression suite in GitHub Actions CI; traced and fixed a <a href="#">40</a>-minute production outage to a missed migration</li></ul>          |

Core Skills

|                       |                                                                                                                                                                       |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AGENTS & RETRIEVAL    | AI agents and agentic workflows; tool use / function calling; planning and evaluation frameworks; RAG with embeddings and vector / semantic search                    |
| LLM APPLICATIONS      | Claude API / Agent SDK, OpenAI and Qwen / Gemini APIs with multi-provider abstraction; prompt engineering; structured outputs with JSON Schema and Zod; SSE streaming |
| EVALS & OBSERVABILITY | evaluation of LLM / agent answers with test question sets; human-in-the-loop review; retries, timeouts, batching; token, cost and quota control                       |
| INTEGRATIONS          | REST APIs; HMAC-SHA1 signed requests; third-party systems (cloud APIs, document storage); AI-assisted development with Claude Code, run headless with tools disabled  |
| ENGINEERING           | Python, JavaScript / TypeScript, Node.js, SQL; Bash and PowerShell; Next.js, React, Express, Flask; Prisma, SQLite, MySQL, PostgreSQL                                 |
| TESTING & DELIVERY    | Vitest, pytest, Jest; GitHub Actions CI/CD; Git / GitHub; Docker; Render, Vercel, Cloudflare Pages / Workers / D1                                                     |

Education

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|-----------------------------------------------------------------------|---------------------|
| Lincoln University                                                    | Nov 2023 – Nov 2024 |
| Master of Applied Computing · Software & Database Development and GIS |                     |
| Zhejiang University of Science & Technology                           | Sep 2006 – Jun 2010 |
| Bachelor of Engineering · Electrical Engineering and Automation       |                     |

Interests

- Cooking — cooking for my family and trying new recipes
- Badminton — footwork, tactics and the people I play with
- Film — how directors tell a story scene by scene